Find the limits in Exercises 9–12.

9. $\lim_{x\to\infty} \frac{\sin 2x}{x}$

10. $\lim_{\theta\to-\infty} \frac{\cos\theta}{3\theta}$

11. $\lim_{t\to-\infty} \frac{2 - t + \sin t}{t + \cos t}$

12. $\lim_{r\to\infty} \frac{r + \sin r}{2r + 7 - 5\sin r}$

Limits of Rational Functions

In Exercises 13–22, find the limit of each rational function **(a)** as $x \to \infty$ and **(b)** as $x \to -\infty$.

13. $f(x) = \frac{2x + 3}{5x + 7}$

14. $f(x) = \frac{2x^3 + 7}{x^3 - x^2 + x + 7}$

15. $f(x) = \frac{x + 1}{x^2 + 3}$

16. $f(x) = \frac{3x + 7}{x^2 - 2}$

17. $h(x) = \frac{7x^3}{x^3 - 3x^2 + 6x}$

18. $h(x) = \frac{9x^4 + x}{2x^4 + 5x^2 - x + 6}$

19. $g(x) = \frac{10x^5 + x^4 + 31}{x^6}$

20. $g(x) = \frac{x^3 + 7x^2 - 2}{x^2 - x + 1}$

21. $f(x) = \frac{3x^7 + 5x^2 - 1}{6x^3 - 7x + 3}$

22. $h(x) = \frac{5x^8 - 2x^3 + 9}{3 + x - 4x^5}$

Divide numerator and denominator by the highest power of x in the denominator and proceed from there. Find the limits in Exercises 23–36.

23. $\lim_{x\to\infty} \sqrt{\dfrac{8x^2-3}{2x^2+x}}$

24. $\lim_{x\to-\infty} \left(\dfrac{x^2+x-1}{8x^2-3}\right)^{1/3}$

25. $\lim_{x\to-\infty} \left(\dfrac{1-x^3}{x^2+7x}\right)^5$

26. $\lim_{x\to\infty} \sqrt{\dfrac{x^2-5x}{x^3+x-2}}$

27. $\lim_{x\to\infty} \dfrac{2\sqrt{x}+x^{-1}}{3x-7}$

28. $\lim_{x\to\infty} \dfrac{2+\sqrt{x}}{2-\sqrt{x}}$

29. $\lim_{x\to-\infty} \dfrac{\sqrt[3]{x}-\sqrt[5]{x}}{\sqrt[3]{x}+\sqrt[5]{x}}$

30. $\lim_{x\to\infty} \dfrac{x^{-1}+x^{-4}}{x^{-2}-x^{-3}}$

31. $\lim_{x\to\infty} \dfrac{2x^{5/3}-x^{1/3}+7}{x^{8/5}+3x+\sqrt{x}}$

32. $\lim_{x\to-\infty} \dfrac{\sqrt[3]{x}-5x+3}{2x+x^{2/3}-4}$

33. $\lim_{x\to\infty} \dfrac{\sqrt{x^2+1}}{x+1}$

34. $\lim_{x\to-\infty} \dfrac{\sqrt{x^2+1}}{x+1}$

35. $\lim_{x\to\infty} \dfrac{x-3}{\sqrt{4x^2+25}}$

36. $\lim_{x\to-\infty} \dfrac{4-3x^3}{\sqrt{x^6+9}}$

Infinite Limits

Find the limits in Exercises 37–48.

37. $\lim_{x \to 0^+} \frac{1}{3x}$

38. $\lim_{x \to 0^-} \frac{5}{2x}$

39. $\lim_{x \to 2^-} \frac{3}{x - 2}$

40. $\lim_{x \to 3^+} \frac{1}{x - 3}$

41. $\lim_{x \to -8^+} \frac{2x}{x + 8}$

42. $\lim_{x \to -5^-} \frac{3x}{2x + 10}$

43. $\lim_{x \to 7} \frac{4}{(x - 7)^2}$

44. $\lim_{x \to 0} \frac{-1}{x^2(x + 1)}$

45. **a.** $\lim_{x \to 0^+} \frac{2}{3x^{1/3}}$ **b.** $\lim_{x \to 0^-} \frac{2}{3x^{1/3}}$

46. **a.** $\lim_{x \to 0^+} \frac{2}{x^{1/5}}$ **b.** $\lim_{x \to 0^-} \frac{2}{x^{1/5}}$

47. $\lim_{x \to 0} \frac{4}{x^{2/5}}$

48. $\lim_{x \to 0} \frac{1}{x^{2/3}}$

Graphing Simple Rational Functions

Graph the rational functions in Exercises 63–68. Include the graphs and equations of the asymptotes and dominant terms.

63. $y = \dfrac{1}{x - 1}$

64. $y = \dfrac{1}{x + 1}$

65. $y = \dfrac{1}{2x + 4}$

66. $y = \dfrac{-3}{x - 3}$

67. $y = \dfrac{x + 3}{x + 2}$

68. $y = \dfrac{2x}{x + 1}$

Domains, Ranges, and Asymptotes

Determine the domain and range of each function. Use various limits to find the asymptotes and the ranges.

69. $y = 4 + \dfrac{3x^2}{x^2 + 1}$

70. $y = \dfrac{2x}{x^2 - 1}$

71. $y = \dfrac{\sqrt{x^2 + 4}}{x}$

72. $y = \dfrac{x^3}{x^3 - 8}$

Inventing Graphs and Functions

In Exercises 73–76, sketch the graph of a function $y = f(x)$ that satisfies the given conditions. No formulas are required—just label the coordinate axes and sketch an appropriate graph. (The answers are not unique, so your graphs may not be exactly like those in the answer section.)

73. $f(0) = 0, f(1) = 2, f(-1) = -2, \lim_{x \to -\infty} f(x) = -1$, and $\lim_{x \to \infty} f(x) = 1$

74. $f(0) = 0, \lim_{x \to \pm\infty} f(x) = 0, \lim_{x \to 0^+} f(x) = 2$, and $\lim_{x \to 0^-} f(x) = -2$

75. $f(0) = 0, \lim_{x \to \pm\infty} f(x) = 0, \lim_{x \to 1^-} f(x) = \lim_{x \to -1^+} f(x) = \infty$, $\lim_{x \to 1^+} f(x) = -\infty$, and $\lim_{x \to -1^-} f(x) = -\infty$

76. $f(2) = 1, f(-1) = 0, \lim_{x \to \infty} f(x) = 0, \lim_{x \to 0^+} f(x) = \infty$, $\lim_{x \to 0^-} f(x) = -\infty$, and $\lim_{x \to -\infty} f(x) = 1$

Finding Limits of Differences When $x \to \pm\infty$

Find the limits in Exercises 84–90. (*Hint*: Try multiplying and dividing by the conjugate.)

84. $\lim_{x\to\infty} (\sqrt{x+9} - \sqrt{x+4})$

85. $\lim_{x\to\infty} (\sqrt{x^2+25} - \sqrt{x^2-1})$

86. $\lim_{x\to-\infty} (\sqrt{x^2+3} + x)$

87. $\lim_{x\to-\infty} (2x + \sqrt{4x^2+3x-2})$

88. $\lim_{x\to\infty} (\sqrt{9x^2-x} - 3x)$

89. $\lim_{x\to\infty} (\sqrt{x^2+3x} - \sqrt{x^2-2x})$

90. $\lim_{x\to\infty} (\sqrt{x^2+x} - \sqrt{x^2-x})$

Oblique Asymptotes

Graph the rational functions in Exercises 103–108. Include the graphs and equations of the asymptotes.

103. $y = \dfrac{x^2}{x - 1}$

104. $y = \dfrac{x^2 + 1}{x - 1}$

105. $y = \dfrac{x^2 - 4}{x - 1}$

106. $y = \dfrac{x^2 - 1}{2x + 4}$

107. $y = \dfrac{x^2 - 1}{x}$

108. $y = \dfrac{x^3 + 1}{x^2}$

Horizontal and Vertical Asymptotes

47. Use limits to determine the equations for all vertical asymptotes.

a. $y = \dfrac{x^2 + 4}{x - 3}$ **b.** $f(x) = \dfrac{x^2 - x - 2}{x^2 - 2x + 1}$

c. $y = \dfrac{x^2 + x - 6}{x^2 + 2x - 8}$

48. Use limits to determine the equations for all horizontal asymptotes.

a. $y = \dfrac{1 - x^2}{x^2 + 1}$ **b.** $f(x) = \dfrac{\sqrt{x} + 4}{\sqrt{x + 4}}$

c. $g(x) = \dfrac{\sqrt{x^2 + 4}}{x}$ **d.** $y = \sqrt{\dfrac{x^2 + 9}{9x^2 + 1}}$

49. Determine the domain and range of $y = \dfrac{\sqrt{16 - x^2}}{x - 2}$.

50. Assume that constants a and b are positive. Find equations for all horizontal and vertical asymptotes for the graph of $y = \dfrac{\sqrt{ax^2 + 4}}{x - b}$.

3

Derivatives

3.1 Tangent Lines and the Derivative at a Point

DEFINITIONS The **slope of the curve** $y = f(x)$ at the point $P(x_0, f(x_0))$ is the number

$$\lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} \qquad \text{(provided the limit exists).}$$

The **tangent line** to the curve at P is the line through P with this slope.

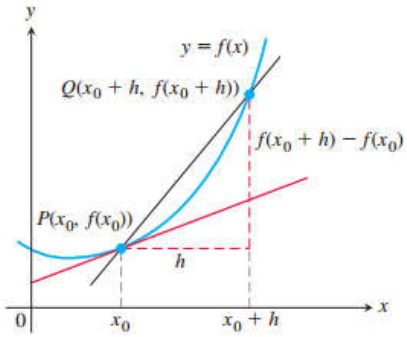
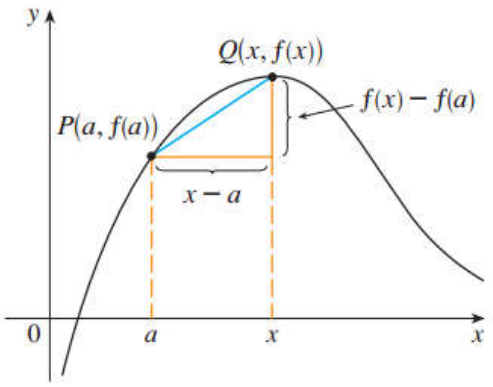


FIGURE 3.1 The slope of the tangent line at P is $\lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$.

1 DEFINITION The **tangent line** to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with slope

$$m = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

provided that this limit exists.



EXAMPLE 2 Find an equation of the tangent line to the hyperbola $y = 3/x$ at the point $(3, 1)$.

Rates of Change: Derivative at a Point

The expression

$$\frac{f(x_0 + h) - f(x_0)}{h}, \quad h \neq 0$$

is called the **difference quotient of f at x_0 with increment h**. If the difference quotient has a limit as h approaches zero, that limit is given a special name and notation.

DEFINITION The **derivative of a function f at a point x_0**, denoted $f'(x_0)$, is

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

provided this limit exists.

5

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

EXAMPLE 4 Find the derivative of the function $f(x) = x^2 - 8x + 9$ at the number a.

The tangent line to $y = f(x)$ at $(a, f(a))$ is the line through $(a, f(a))$ whose slope is equal to $f'(a)$, the derivative of f at a.

If we use the point-slope form of the equation of a line, we can write an equation of the tangent line to the curve $y = f(x)$ at the point $(a, f(a))$:

$$y - f(a) = f'(a)(x - a)$$

EXAMPLE 5 Find an equation of the tangent line to the parabola $y = x^2 - 8x + 9$ at the point $(3, -6)$.

Summary

We have been discussing slopes of curves, lines tangent to a curve, the rate of change of a function, and the derivative of a function at a point. All of these ideas are based on the same limit.

The following are all interpretations for the limit of the difference quotient

$$\lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

1. The slope of the graph of $y = f(x)$ at $x = x_0$
2. The slope of the tangent line to the curve $y = f(x)$ at $x = x_0$
3. The rate of change of $f(x)$ with respect to x at the $x = x_0$
4. The derivative $f'(x_0)$ at $x = x_0$

Notation

There are many ways to denote the derivative of a function $y = f(x)$, where the independent variable is x and the dependent variable is y. Some common alternative notations for the derivative include

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x) = D(f)(x) = D_x f(x).$$

The symbols d/dx and D indicate the operation of differentiation. We read dy/dx as "the derivative of y with respect to x," and df/dx and $(d/dx)f(x)$ as "the derivative of f with respect to x." The "prime" notations y' and f' originate with Newton. The d/dx notations are similar to those used by Leibniz. The symbol dy/dx should not be regarded as a ratio; it is simply a notation that denotes a derivative.

To indicate the value of a derivative at a specified number $x = a$, we use the notation

$$f'(a) = \left.\frac{dy}{dx}\right|_{x=a} = \left.\frac{df}{dx}\right|_{x=a} = \left.\frac{d}{dx}f(x)\right|_{x=a}.$$

RATES OF CHANGE

Suppose y is a quantity that depends on another quantity x. Thus y is a function of x and we write $y = f(x)$. If x changes from x_1 to x_2, then the change in x (also called the **increment** of x) is

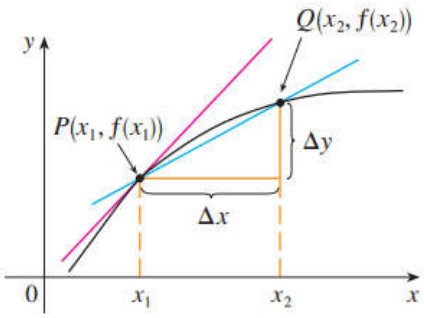
$$\Delta x = x_2 - x_1$$

and the corresponding change in y is

$$\Delta y = f(x_2) - f(x_1)$$

The difference quotient

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$



average rate of change $= m_{PQ}$

instantaneous rate of change $=$ slope of tangent at P

6 instantaneous rate of change $= \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{x_2 \to x_1} \frac{f(x_2) - f(x_1)}{x_2 - x_1}$

We recognize this limit as being the derivative $f'(x_1)$.

We know that one interpretation of the derivative $f'(a)$ is as the slope of the tangent line to the curve $y = f(x)$ when $x = a$. We now have a second interpretation:

The derivative $f'(a)$ is the instantaneous rate of change of $y = f(x)$ with respect to x when $x = a$.

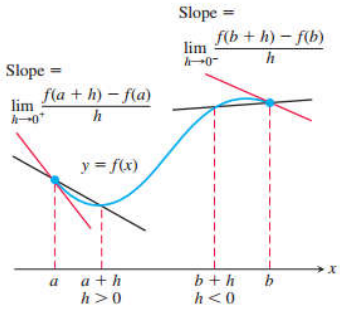


FIGURE 3.7 Derivatives at endpoints of a closed interval are one-sided limits.

Differentiable on an Interval; One-Sided Derivatives

A function $y = f(x)$ is **differentiable on an open interval** (finite or infinite) if it has a derivative at each point of the interval. It is **differentiable on a closed interval** $[a, b]$ if it is differentiable on the interior (a, b) and if the limits

$$\lim_{h\to 0^+} \frac{f(a+h)-f(a)}{h} \qquad \textbf{Right-hand derivative at } a$$

$$\lim_{h\to 0^-} \frac{f(b+h)-f(b)}{h} \qquad \textbf{Left-hand derivative at } b$$

exist at the endpoints (Figure 3.7).

Right-hand and left-hand derivatives may or may not be defined at any point of a function's domain. Because of Theorem 6, Section 2.4, a function has a derivative at an interior point if and only if it has left-hand and right-hand derivatives there, and these one-sided derivatives are equal.

EXAMPLE 4 Show that the function $y = |x|$ is differentiable on $(-\infty, 0)$ and on $(0, \infty)$ but has no derivative at $x = 0$.

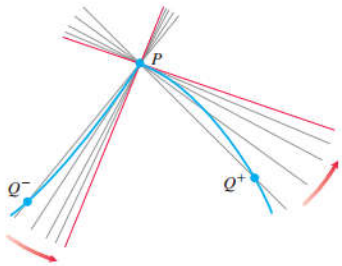
EXAMPLE 5 In Example 2 we found that for $x > 0$,

$$\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}.$$

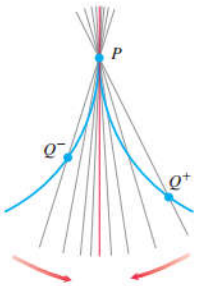
l tangent

When Does a Function *Not* Have a Derivative at a Point?

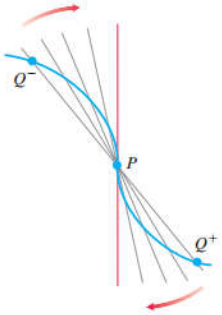
A function has a derivative at a point x_0 if the slopes of the secant lines through $P(x_0, f(x_0))$ and a nearby point Q on the graph approach a finite limit as Q approaches P. Thus differentiability is a "smoothness" condition on the graph of f. A function can fail to have a derivative at a point for many reasons, including the existence of points where the graph has



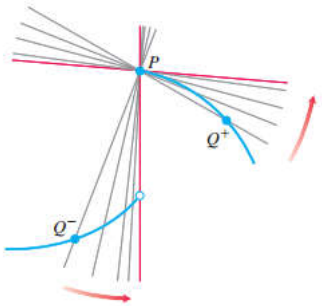
1. a *corner*, where the one-sided derivatives differ

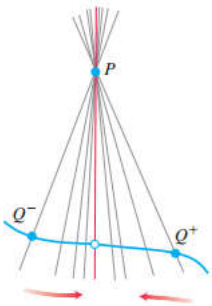


2. a *cusp*, where the slope of PQ approaches ∞ from one side and $-\infty$ from the other

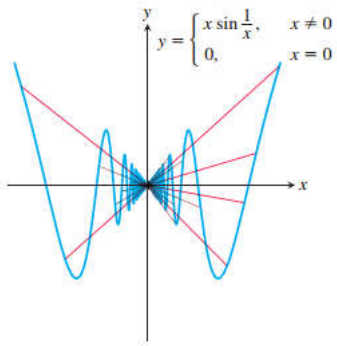


3. a *vertical tangent line*, where the slope of PQ approaches ∞ from both sides or approaches $-\infty$ from both sides (here, $-\infty$)





4. a *discontinuity* (two examples shown)



5. wild oscillation

Differentiable Functions Are Continuous

A function is continuous at every point where it has a derivative.

THEOREM 1—Differentiability Implies Continuity If f has a derivative at $x = c$, then f is continuous at $x = c$.

EXERCISES 3.2

Finding Derivative Functions and Values

Using the definition, calculate the derivatives of the functions in Exercises 1–6. Then find the values of the derivatives as specified.

1. $f(x) = 4 - x^2$; $f'(-3), f'(0), f'(1)$
2. $F(x) = (x - 1)^2 + 1$; $F'(-1), F'(0), F'(2)$
3. $g(t) = \frac{1}{t^2}$; $g'(-1), g'(2), g'(\sqrt{3})$
4. $k(z) = \frac{1 - z}{2z}$; $k'(-1), k'(1), k'(\sqrt{2})$
5. $p(\theta) = \sqrt{3\theta}$; $p'(1), p'(3), p'(2/3)$
6. $r(s) = \sqrt{2s + 1}$; $r'(0), r'(1), r'(1/2)$

In Exercises 7–12, find the indicated derivatives.

7. $\frac{dy}{dx}$ if $y = 2x^3$
8. $\frac{dr}{ds}$ if $r = s^3 - 2s^2 + 3$
9. $\frac{ds}{dt}$ if $s = \frac{t}{2t + 1}$
10. $\frac{dv}{dt}$ if $v = t - \frac{1}{t}$
11. $\frac{dp}{dq}$ if $p = q^{3/2}$
12. $\frac{dz}{dw}$ if $z = \frac{1}{\sqrt{w^2 - 1}}$

Slopes and Tangent Lines

In Exercises 13–16, differentiate the functions and find the slope of the tangent line at the given value of the independent variable.

13. $f(x) = x + \frac{9}{x}, \quad x = -3$

14. $k(x) = \frac{1}{2 + x}, \quad x = 2$

15. $s = t^3 - t^2, \quad t = -1$

16. $y = \frac{x + 3}{1 - x}, \quad x = -2$

In Exercises 17–18, differentiate the functions. Then find an equation of the tangent line at the indicated point on the graph of the function.

17. $y = f(x) = \frac{8}{\sqrt{x - 2}}, \quad (x, y) = (6, 4)$

18. $w = g(z) = 1 + \sqrt{4 - z}, \quad (z, w) = (3, 2)$

In Exercises 19–22, find the values of the derivatives.

19. $\left.\frac{ds}{dt}\right|_{t=-1}$ if $s = 1 - 3t^2$

20. $\left.\frac{dy}{dx}\right|_{x=\sqrt{3}}$ if $y = 1 - \frac{1}{x}$

21. $\left.\frac{dr}{d\theta}\right|_{\theta=0}$ if $r = \frac{2}{\sqrt{4 - \theta}}$

22. $\left.\frac{dw}{dz}\right|_{z=4}$ if $w = z + \sqrt{z}$

Using the Alternative Formula for Derivatives

Use the formula

$$f'(x) = \lim_{z \to x} \frac{f(z) - f(x)}{z - x}$$

to find the derivative of the functions in Exercises 23–26.

23. $f(x) = \dfrac{1}{x + 2}$

24. $f(x) = x^2 - 3x + 4$

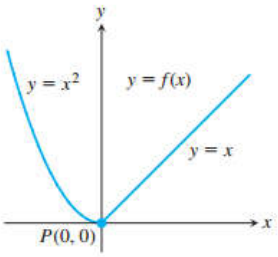
25. $g(x) = \dfrac{x}{x - 1}$

26. $g(x) = 1 + \sqrt{x}$

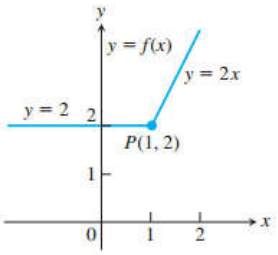
One-Sided Derivatives

Compute the right-hand and left-hand derivatives as limits to show that the functions in Exercises 37–40 are not differentiable at the point P.

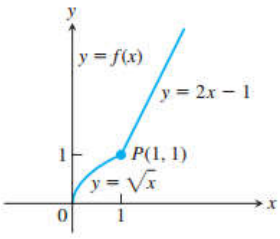
37.



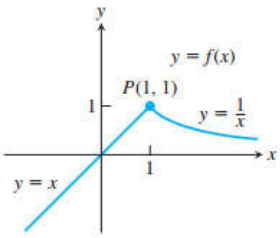
38.



39.



40.



EXAMPLE 6 A manufacturer produces bolts of a fabric with a fixed width. The cost of producing x yards of this fabric is $C = f(x)$ dollars.

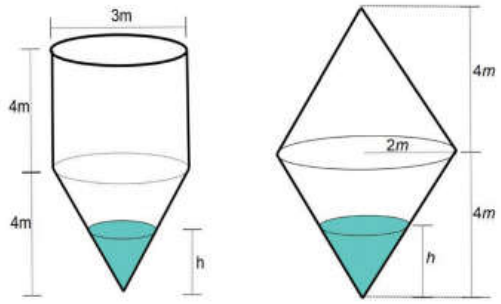
(a) What is the meaning of the derivative $f'(x)$? What are its units?

(b) In practical terms, what does it mean to say that $f'(1000) = 9$?

(c) Which do you think is greater, $f'(50)$ or $f'(500)$? What about $f'(5000)$?

Modelling activity: the water tank

Two different shaped water tanks are seen below. These tanks are being filled with water at a constant flow rate. Sketch the graph of change in volume with respect to height and write a direction explaining how you draw it.



Answer the following questions:

(a) Does the relation between volume and height define a function? Discuss by drawing the graph.
(b) By looking at your graph, decide if derivative exists or not at the transition points of the water tanks. For example, does derivative exist at the transition point from the conical part to the cylindrical part for Tank-1? Explain your reasoning.
(c) Can the relationship between volume and height be modelled with an algebraic equation or function? If possible, try to find an algebraic model.
(d) Test the result that you obtained in part-b with the algebraic model you reached in part-c. What do you get?

1. **(a)** Find the value of a that makes the function

$$f(x) = \begin{cases} \frac{(1+x)^n - 1}{x} & \text{if } x \neq 0 \\ a & \text{if } x = 0 \end{cases}.$$

continous at x=0, where n is a positive integer.

(b) Find $\lim \lim_{x \to \circ} \dfrac{6x - \sin 2x}{2x + 3\sin 4x}$, if it exists.

2. Consider the function $f(x) = \dfrac{2x^2 + 5x + 1}{x - 1}$

   (a) What is the domain of f ?

   (b) Find all horizontal, vertical and oblique (slant) asymptotes.

   (c) Find the interval(s) on which f is increasing or decreasing.

3. **(a)** Let

$$f(x) = \begin{cases} x^2 + 2 & \text{if } x \neq 0 \\ 4x - 2 & \text{if } x = 0 \end{cases}.$$

Determine whether f has a tangent line at the point P(2,6); if so, find an equation of the tangent line.

1. Evaluate the following limits :

   (a) $\lim_{x \to 0} \frac{\tan x}{\sqrt{1 - \cos x}}$.

(5) *Consider the function* $f(x) = x \cdot |x|$.

a) *Is* f *differentiable everywhere? Justify your answer.*

